

# Pritam Mahanta

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Leetcode | Codeforces | Codechef | GitHub

## EDUCATION

### Jalpaiguri Government Engineering College

- Bachelor of Technology in Information Technology

September 2023 – Expected May 2027

CGPA: 8.5

## EXPERIENCE

### Member of Technical Staff Intern

GeeksforGeeks

March 2026 – May 2026

Greater Noida, UP

- Validated cross-language code translations in **5+ languages** for logic, time complexity, and clean formatting, reducing manual editorial review effort by **50%**.
- Integrated internal code-execution tools to test edge cases in real-time, streamlining the problem validation process and cutting debugging cycles to under **3 seconds**.
- Authored and tested **20+ DSA** problems with comprehensive edge-case test suites for coding contests, serving an active user base of **50,000+** developers.

## PROJECTS

### Relay – Distributed Job Queue System

August 2026

NestJS, TypeScript, PostgreSQL, Redis, BullMQ, Docker | GitHub

- Engineered a **backend job-processing system** with **BullMQ, Redis, and PostgreSQL**, using independent workers with configurable concurrency, **4 priority levels**, delayed execution, and persistent job state.
- Implemented **exponential-backoff retries**, **dead-letter queue (DLQ)**, idempotent job creation, and **60s job execution timeouts**, containerized the **API, worker, Redis, and PostgreSQL** with Docker Compose.

### Psh – POSIX-Compliant Unix Shell in C

January 2026

C, POSIX APIs, Process and Job Control, Signals, I/O Redirection | GitHub

- Architected a **POSIX-Compliant Unix shell in C** with process-group job control (`setpgid`, `tcsetpgrp`), signal forwarding via `sigaction`, and non-blocking process reaping with `WNOHANG`.
- Built a **recursive-descent parser** for multi-stage pipelines and I/O redirection (`pipe`, `dup2`), with per-stage file-descriptor management to prevent descriptor leaks.
- Profiled **10,000 command runs**, measuring **537  $\mu$ s median / 774  $\mu$ s p95** execution latency and **7.28  $\mu$ s median / 14.68  $\mu$ s p95** job-control overhead.

### CF Performance Mirror – Browser Extension

October 2025

React, TypeScript, Manifest V3, Web Extensions, REST APIs | Chrome | Firefox | Github

- Published a **React + TypeScript** Browser extension that analyzes Codeforces submissions, contest history, and rating changes across 5 divisions, with 3 analysis modes and configurable time- and contest-based ranges.
- Scaled browser extension to **800+ installs**, **450+ daily active users**, and a **4.9/5 rating** across **237000+** store impressions by implementing client-side caching via `chrome.storage` to optimize API request overhead.

## ACHIEVEMENTS AND AWARDS

- Secured **Rank 57** at the 2025 **ICPC Kanpur Regional** onsite round.
- Ranked 680** in India and **2320 globally** at **Meta Hacker Cup 2025** (Round 1).
- Finished **51st globally** out of **10,000+** participants at CodeChef Starters 159 (2025).
- Placed 9th at the Jadavpur TechFest Srijan 2025 finals (ICPC-style), advancing via All-India prelims.
- Competitive Programming: Codeforces** (Specialist, **1552**), **CodeChef** (4\*, **1800+**), **LeetCode** (Knight, **1900+**), alongside active participation on CSES.

Certificate

## TECHNICAL SKILLS

**Languages:** C++, C, JavaScript, TypeScript, SQL

**Web Technologies:** Node.js, Express.js, NestJS, React.js, Tailwind CSS, RESTful APIs

**Databases and Messaging:** Redis, MongoDB, PostgreSQL, MySQL, BullMQ

**Tools:** Linux, Bash, Docker, Docker Compose, Git, GitHub Actions

**Core Competencies:** Data Structures and Algorithms, Operating Systems, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Computer Networks

## LEADERSHIP & MENTORSHIP

- Event Lead, JICPC (Sristi Tech Fest):** Led organization and problem-setting for an ICPC-style programming contest, coordinating execution for **30+ teams** across multiple colleges.